



GIRRAWEE HIGH SCHOOL
TASK 1 2021 (*December 2020*)

MATHEMATICS

EXTENSION 1

Time allowed – 90 Minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Each question attempted is to be marked clearly Question 6, Question 7 etc. Each question is to be returned on a separate page in your answer booklet.
- You may ask for spare answer booklets if you need them.
- For Multiple choice: Fill in the circle corresponding to the correct answer on the multiple choice answer sheet in your answer booklet.

For questions 1-5, fill in the response oval corresponding to the correct answer on your Multiple-choice answer sheet.

(1) Find the exact value of $\cos 35^\circ \cos 5^\circ + \sin 35^\circ \sin 5^\circ$

- A) $\frac{1}{2}$ B) $-\frac{1}{2}$ C) $\frac{\sqrt{3}}{2}$ D) $-\frac{\sqrt{3}}{2}$

(2) Given that $4\sin\theta + 3\cos\theta = -5$, what is the value of $\tan\frac{\theta}{2}$

- A) -2 B) -1 C) 1 D) 2

(3) What is the value of $\sin 2\theta$ given that $\sin\theta = \frac{\sqrt{3}}{4}$ and θ is obtuse?

- A) $-\frac{\sqrt{39}}{8}$ B) $-\frac{\sqrt{3}}{2}$ C) $\frac{\sqrt{3}}{2}$ D) $\frac{\sqrt{39}}{8}$

(4) What is the coefficient of the x^3 term in the expansion of $(2 - 3x)^8$?

- A) 56 B) -1512 C) 1512 D) -48384

(5) $2\cos 6\theta \sin 2\theta$ can be written as

- A) $\cos 4\theta - \cos 8\theta$ B) $\cos 8\theta + \cos 4\theta$ C) $\sin 8\theta - \sin 4\theta$ D) $\sin 8\theta + \sin 4\theta$

Question 6 (19 marks)

- a) Given that the angles A and B are acute, and that $\sin A = \frac{3}{5}$ and $\cos B = \frac{5}{13}$,
find the value of $\sin(A - B)$. 3
- b) Use compound angle formulae to find the exact value of
- i) $2\sin 15^\circ \cos 15^\circ$ 2
- ii) $\cos 105^\circ$ 3
- c) Given that $\sin A = \frac{2}{3}$, where $\frac{\pi}{2} < A < \pi$ and $\tan B = \frac{2}{3}$ where $\pi < B < \frac{3\pi}{2}$,
show that $\cos(A + B) = \frac{3\sqrt{5} + 4}{3\sqrt{13}}$ 4
- d) Show that $\tan 35^\circ + \tan 10^\circ + \tan 35^\circ \tan 10^\circ = 1$ 4
- e) By writing 3θ as $2\theta + \theta$ and using compound-angle and double-angle results,
Prove that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$. 3

Question 7 (16 marks)

- a) Eliminate θ from the following pair of parametric equations.
- $$x = 2 + \cos \theta \quad \text{and} \quad y = \cos 2\theta$$
- 3
- b) Use the products– to– sums identities to show that
- i) $2\sin 70^\circ \sin 50^\circ = \frac{1}{2} + \cos 20^\circ$ 2
- ii) $\frac{\sin 3\theta + \sin \theta}{\cos 3\theta + \cos \theta} = \tan 2\theta$ 4
- c) Prove the identities:
- i) $\frac{1}{1 - \tan \theta} - \frac{1}{1 + \tan \theta} = \tan 2\theta$ 3
- ii) $4\sin 5\theta \cos 3\theta \sin \theta = \cos \theta - \cos 3\theta + \cos 7\theta - \cos 9\theta$ 4

Question 8 (15marks)

- a) Expand and simplify $\left(\frac{2}{x} + 3x^2\right)^5$ 2
- b) Find the term in $x^{10}y^2$ in the expansion of $\left(\frac{1}{2}x - 3y^2\right)^{11}$ 3
- c) i) Expand $(1+7x)^5$ as far as the term in x^2 . 1
- ii) Hence find the coefficient of x^2 in the expansion of $(1-5x)(1+7x)^5$. 3
- d) Find the term independent of x in the expansion of $(2x + x^{-2})^6$. 3
- e) In the expansion of $(2+3x)^n$, the coefficients of x^5 and x^5 are in the ratio 4:9
Find the value of n . 3

Question 9 (18marks)

- a) Using an appropriate substitution in the expansion of $(1+x)^n$,
show that $\binom{n}{0} + 2 \times \binom{n}{1} + 4 \times \binom{n}{2} + 8 \times \binom{n}{3} + \dots = 3^n$ 3
- b) By expanding and differentiating $(1+x)^{2n}$, show that :
 $1 \times \binom{2n}{1} + 2 \times \binom{2n}{2} + \dots + 2n \times \binom{2n}{2n} = n \times 2^{2n}$ 4
- c) By letting $t = \tan \frac{x}{2}$, solve $\sin x + \cos x = -1$ in the domain $0 \leq x \leq 2\pi$ 4
- d) i) Express $\sqrt{3} \cos \theta - \sin \theta$ in the form $R \cos(\theta + \alpha)$, where $R > 0$ and
 $0 < \alpha < \frac{\pi}{2}$ 4
- ii) Hence or otherwise, solve $\sqrt{3} \cos \theta - \sin \theta = -1$ for $0 \leq \theta \leq 2\pi$ 3

Question 10 (14marks)

- a) Use the principle of mathematical induction to show that for all integers $n \geq 1$,

$$1 \times 2 + 2 \times 5 + 3 \times 8 + \dots + n(3n-1) = n^2(n+1) \quad 3$$

- b) Prove by mathematical induction that $4^n + 14$ is divisible by 6 for all positive integers $n \geq 1$. 4

- c) Solve: $(n+2)! = 72n!$ 3

- d) Use mathematical induction to prove that, for positive integers n ,

$$1 + 3 + 9 + \dots + 3^{n-1} = \frac{1}{2}(3^n - 1) \quad 4$$

.

End of examination!!!

Mathematics Extension 1 Task 1 SOLUTIONS 2020

multiple choice questions

① C

$$\cos(35^\circ - 5^\circ) = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

② A

$$\frac{4 \times 2t}{1+t^2} + \frac{3(1-t^2)}{1+t^2} = -5$$

$$8t + 3 - 3t^2 = -5 - 5t^2$$

$$2t^2 + 8t + 8 = 0 \Rightarrow (t+2)^2 = 0$$

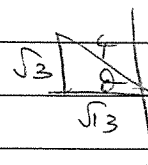
$$t = -2$$

③ A

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$= 2 \times \frac{\sqrt{3}}{4} \times \left(-\frac{\sqrt{13}}{4}\right)$$

$$= -\frac{\sqrt{39}}{8}$$



④ D

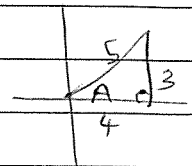
$$= 8 \binom{5}{3} \times 2^5 \times (-3)^3 = -48384$$

⑤ C

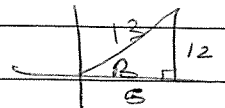
$$2 \cos 6\theta \sin 2\theta = \sin 8\theta - \sin 4\theta$$

Questions

a) $\sin A = \frac{3}{5}$ and $\cos B = \frac{5}{13}$



$$\begin{aligned}\sin(A-B) &= \sin A \cos B - \cos A \sin B \\ &= \frac{3}{5} \times \frac{5}{13} - \frac{4}{5} \times \frac{12}{13} \\ &= \frac{15}{65} - \frac{48}{65} = \frac{-33}{65}\end{aligned}$$



(3)

b) (i) $2 \sin 15^\circ \cos 15^\circ = \sin 30^\circ = \frac{1}{2}$

(1)

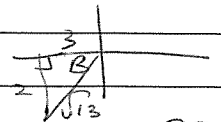
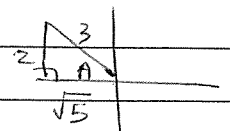
$$\begin{aligned}\text{(ii)} \quad \cos(105^\circ) &= \cos(60^\circ + 45^\circ) = \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ \\ &= \frac{1}{2\sqrt{2}} - \frac{\sqrt{3}}{2\sqrt{2}} = \frac{1-\sqrt{3}}{2\sqrt{2}} \\ &= \frac{\sqrt{2}-\sqrt{6}}{4}\end{aligned}$$

(3)

c) $\cos(A+B) = \cos A \cos B - \sin A \sin B$

$$= \left(-\frac{\sqrt{5}}{3} \times -\frac{3}{\sqrt{13}}\right) - \left(\frac{2}{3} \times \frac{2}{\sqrt{13}}\right)$$

$$= \frac{3\sqrt{5} + 4}{3\sqrt{13}}$$



(4)

d) $\tan 45^\circ = \tan(35^\circ + 10^\circ) = \frac{\tan 35^\circ + \tan 10^\circ}{1 - \tan 35^\circ \tan 10^\circ} = 1$

$$\tan 35^\circ + \tan 10^\circ = 1 - \tan 35^\circ \tan 10^\circ$$

$$\tan 35^\circ + \tan 10^\circ + \tan 35^\circ \tan 10^\circ = 1$$

(4)

Question 6 (continued)

c) LHS

$$\begin{aligned}\cos 3\theta &= \cos(2\theta + \theta) = \cos 2\theta \cos \theta - \sin 2\theta \sin \theta \\&= (2\cos^2 \theta - 1)\cos \theta - 2\sin \theta \cos \theta \sin \theta \\&= 2\cos^3 \theta - \cos \theta - 2\sin^2 \theta \cos \theta \\&= 2\cos^3 \theta - \cos \theta - 2\cos \theta (1 - \cos^2 \theta) \\&= 2\cos^3 \theta - \cos \theta - 2\cos \theta + 2\cos^3 \theta \\&= 4\cos^3 \theta - 3\cos \theta = \text{RHS}\end{aligned}$$

Question 7

(3)

a) $\cos \theta = x - 2$ and $y = \cos 2\theta$.

$$\cos 2\theta = 2\cos^2 \theta - 1$$

$$\therefore 2(x-2)^2 - 1 = y \Rightarrow 2x^2 - 8x + 7$$

$$\therefore y = 2x^2 - 8x + 7$$

(3)

b) LHS

$$(i) 2\sin 70^\circ \sin 50^\circ = \cos 20^\circ - \cos 120^\circ$$

$$= \cos 20^\circ + \frac{1}{2} = \text{RHS} \quad (3)$$

(ii) LHS

$$\begin{aligned}\frac{\sin 3\theta + \sin \theta}{\cos 3\theta + \cos \theta} &= \frac{1}{2} [\sin(2\theta + \theta) + \sin(2\theta - \theta)] \\&= \frac{1}{2} [\cos(2\theta + \theta) + \cos(2\theta - \theta)]\end{aligned}$$

$$A = 2\theta$$

$$B = \theta$$

$$= \frac{\sin 2\theta \cos \theta}{\cos 2\theta \cos \theta} = \tan 2\theta = \text{RHS}$$

(4)

Question 7 (continued).

$$c) \quad (i) \quad \frac{1}{1-\tan\theta} - \frac{1}{1+\tan\theta} = \tan 2\theta$$

$$\begin{aligned} \text{LHS} \quad \frac{1}{1-\tan\theta} - \frac{1}{1+\tan\theta} &= \frac{1+\tan\theta - 1+\tan\theta}{1-\tan^2\theta} \\ &= \frac{2\tan\theta}{1-\tan^2\theta} = \tan 2\theta = \text{RHS} \end{aligned}$$

(3)

$$(ii) \quad 4 \sin 5\theta \cos 3\theta \sin \theta = \cos \theta - \cos 3\theta + \cos 7\theta - \cos 9\theta$$

$$\text{LHS} \quad 4 \sin 5\theta \cos 3\theta \sin \theta = 2 \sin \theta [2 \sin 5\theta \cos 3\theta]$$

$$= 2 \sin \theta [\sin 8\theta + \sin 2\theta]$$

$$= 2 \sin \theta \sin 8\theta + 2 \sin \theta \sin 2\theta$$

$$= \cos 7\theta - \cos 9\theta + \cos \theta - \cos 3\theta$$

$$= \cos \theta - \cos 3\theta + \cos 7\theta - \cos 9\theta = \text{RHS}$$

(4)

Question 8.

$$\begin{aligned} a) \quad \left(\frac{2}{x} + 3x^2\right)^5 &= {}^5C_0 \left(\frac{2}{x}\right)^5 + {}^5C_1 \left(\frac{2}{x}\right)^4 (3x^2) + {}^5C_2 \left(\frac{2}{x}\right)^3 (3x^2)^2 \\ &\quad + {}^5C_3 \left(\frac{2}{x}\right)^2 (3x^2)^3 + {}^5C_4 \left(\frac{2}{x}\right) (3x^2)^4 + {}^5C_5 (3x^2)^5 \end{aligned}$$

$$\begin{aligned} &= \frac{32}{x^5} + \frac{5 \times 48}{x^2} + 10 \times 8 \times 9x + 10 \times 4 \times 27x^4 \\ &\quad + 810x^7 + 243x^{10} \end{aligned}$$

$$= \frac{32}{x^5} + \frac{240}{x^2} + 720x + 1080x^4 + 810x^7 + 243x^{10}$$

(2)

Question 8 (continued).

b) $\left(\frac{1}{2}x - 3y^2\right)^{11}$

$$x^{10}y^2 \text{ term} \Rightarrow {}^{11}C_1 \left(\frac{x}{2}\right)^{10} (-3y^2)^1 = \frac{-33 x^{10} y^2}{2^{10}}$$

$$= \frac{-33}{1024} x^{10} y^2$$

(3)

c) (i) $(1+7x)^5 = {}^5C_0 + {}^5C_1(7x) + {}^5C_2(7x)^2 + \dots$

(1)

$$(ii) (1-5x)({}^5C_0 + {}^5C_1(7x) + {}^5C_2(7x)^2 + \dots)$$

$$= (1-5x)(1 + 35x + 490x^2 + \dots)$$

$$\text{coefficient of } x^2 \Rightarrow 490 - 5 \times 35 = 315$$

(3)

d) $(2x + x^{-2})^6$

$$T_{k+1} = {}^6C_k \cdot (2x)^{6-k} (x^{-2})^k = {}^6C_k \cdot 2^{6-k} \cdot x^{6-3k}$$

$$\text{independent of } x \Rightarrow 6-3k=0$$

$$\therefore k=2$$

$$\therefore T_3 = {}^6C_2 \cdot 2^4 = 15 \times 16 = 240$$

(3)

$$d) (2+3x)^n$$

$$C_5 \text{ coefficient of } x^5 = nC_5 \cdot 2^{n-5} \cdot x^5$$

$$C_6 \text{ coefficient of } x^6 = nC_6 \cdot 2^{n-6} \cdot x^6$$

$$\frac{nC_5 \cdot 2^{n-5} \cdot x^5}{nC_6 \cdot 2^{n-6} \cdot x^6} = \frac{4}{9}$$

$$\frac{n!}{(n-5)!(5!)} \times \frac{6! \cdot (n-6)!}{n!} \times \frac{2}{3} = \frac{4}{9}$$

$$\frac{6}{n-5} \times \frac{2}{3} = \frac{4}{9} = \frac{4}{n-5}$$

$$\therefore 4n - 20 = 36$$

$$4n = 56 \quad \therefore n = 14$$

(3)

Question 9

$$a) (i) (1+x)^n = nC_0 + nC_1 x + nC_2 x^2 + \dots + nC_n x^n \quad (1)$$

1. substitute $x=2$ in the expansion $\Rightarrow$

$$3^n = nC_0 + 2 \times nC_1 + 4 \times nC_2 + \dots \quad (2)$$

$$b) (1+x)^{2n} = {}^{2n}C_0 + {}^{2n}C_1 x + {}^{2n}C_2 x^2 + \dots + {}^{2n}C_{2n-1} x^{2n-1} + {}^{2n}C_{2n} x^{2n}$$

Differentiating both sides,

$$2n(1+x)^{2n-1} = 2nC_1 + 2x {}^{2n}C_2 + \dots + 2n x^{2n-1} {}^{2n}C_{2n}$$

substitute $x=1$ both sides,

$$2n \times 2^{2n-1} = 2nC_1 + 2 \times 2 {}^{2n}C_2 + \dots + 2n \times 2^{2n} {}^{2n}C_{2n}$$

$$\therefore n \times 2^{2n} = 2nC_1 + 2 \times 2 {}^{2n}C_2 + \dots + 2n \times 2^{2n} {}^{2n}C_{2n} \quad (3)$$

$$c) \sin x + \cos x = -1 \quad 0 \leq x < 2\pi$$

$$\text{sub. } t = \tan \frac{x}{2} \Rightarrow \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = -1$$

$$\therefore 2t + 1 - t^2 = -1 - t^2 \Rightarrow 2t = -2 \\ t = -1$$

$$\therefore \tan \frac{x}{2} = -1 \Rightarrow \frac{x}{2} = \frac{3\pi}{4} \therefore x = \frac{3\pi}{2}$$

check $x = \pi \Rightarrow$ LHS $\sin \pi + \cos \pi = 0 + (-1) = -1 = \text{RHS}$

$\therefore x = \pi$ is a solution $\therefore x = \frac{3\pi}{2}, \pi$ (3)

$$d) (i) \sqrt{3} \cos \theta - \sin \theta = R \cos(\theta + \alpha)$$

$$\sqrt{3} \cos \theta - \sin \theta = R \cos \theta \cos \alpha - R \sin \theta \sin \alpha$$

equating coefficients of $\cos \theta$,

$$R \cos \alpha = \sqrt{3} \quad \text{--- (1)}$$

$$R \sin \alpha = 1 \quad \text{--- (2)}$$

$$\textcircled{1}^2 + \textcircled{2}^2 \Rightarrow R^2 = 4 \quad \therefore R = 2 \quad (R > 0)$$

$$\sin \alpha = \frac{1}{2} \quad \text{and} \quad \cos \alpha = \frac{\sqrt{3}}{2}$$

$\therefore \alpha$ is in quadrant I and $\alpha = \frac{\pi}{6}$

$$\therefore \sqrt{3} \cos \theta - \sin \theta = 2 \cos\left(\theta + \frac{\pi}{6}\right)$$

$$(ii) \quad 2 \cos\left(\theta + \frac{\pi}{6}\right) = -1$$

$$\therefore \cos\left(\theta + \frac{\pi}{6}\right) = -\frac{1}{2}$$

$$\therefore \theta + \frac{\pi}{6} = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$0 \leq \theta < 2\pi$$

$$\frac{2\pi}{6} \leq \theta + \frac{\pi}{6} < \frac{13\pi}{6}$$

$$\therefore \theta = \frac{2\pi}{3} - \frac{\pi}{6} \quad \text{or} \quad \frac{4\pi}{3} - \frac{\pi}{6}$$

$$= \frac{\pi}{2} \quad \text{or} \quad \frac{7\pi}{6}$$

Question 10

a) $1 \times 2 + 2 \times 5 + 3 \times 8 + \dots + n(3n-1) = n^2(n+1)$ for $n \geq 1$

step 1 prove true for $n=1$

$$\text{LHS } 1 \times 2 = 2 = 1^2 \times 2 = 2 = \text{RHS}$$

$\therefore$ true for $n=1$

step 2 Assume true for $n=k$

$$\text{i.e. } 1 \times 2 + 2 \times 5 + \dots + k(3k-1) = k^2(k+1) \quad (1)$$

step 3 prove true for $n=k+1$

$$\text{i.e. } 1 \times 2 + 2 \times 5 + \dots + k(3k-1) + (k+1)(3k+2) = (k+1)^2(k+2)$$

$$\text{LHS } 1 \times 2 + 2 \times 5 + \dots + k(3k-1) + (k+1)(3k+2)$$

$$= k^2(k+1) + (k+1)(3k+2) \quad \text{from (1)}$$

$$= (k+1)(k^2 + 3k + 2) = (k+1)(k+1)(k+2)$$

$$= (k+1)^2(k+2) = \text{RHS}$$

$\therefore$ true for $n=k+1$

step 4:

Since true for $n=1$, by PMI, true for all integers $n \geq 1$

b) $4^n + 14$ is divisible by 6 for $n \geq 1$.

step 1 : show true for $n=1$

$$4^1 + 14 = 18 = 6 \times 3 \therefore \text{true for } n=1$$

step 2 Assume true for $n=k$

$$\therefore 4^k + 14 = 6P \quad (P = \text{integer}) \quad \text{--- (1)}$$

step 3 Prove true for $n=k+1$.

$$\therefore 4^{k+1} + 14 = 6Q \quad (Q = \text{integer}).$$

$$\text{LHS} \quad 4^{k+1} + 14 = 4 \times 4^k + 14$$

$$= 4(6P - 14) - 14$$

$$= 24P - 42 = 6(4P - 7)$$

$$= 6Q, \quad Q = \text{integer}$$

$\therefore$ true for $n=k+1$.

step 4

Since true for $n=1$, by PMI, true for all $n \geq 1$.

$$c) (n+2)! = 72n!$$

$$(n+2)(n+1) \times n! = 72n!$$

$$\therefore (n+2)(n+1) = 72 = n^2 + 3n + 2$$

$$\therefore n^2 + 3n - 70 = 0$$

$$(n+10)(n-7) = 0 \quad \therefore n = 7 \quad (\because n \geq 0).$$

$$d) \quad 1 + 3 + 9 + \dots + 3^{n-1} = \frac{1}{2}(3^n - 1)$$

for positive integers

$n \geq 1$

step 1 Prove true for $n=1$

$$\text{LHS} = 1$$

$$\text{RHS} = \frac{1}{2}(3-1) = 1$$

$\therefore$ true for $n=1$

step 2 Assume true for $n=k$

$$\text{ie } 1 + 3 + 9 + \dots + 3^{k-1} = \frac{1}{2}(3^k - 1) \quad (1)$$

step 3 Prove true for $n=k+1$.

$$\text{ie } 1 + 3 + 9 + \dots + 3^{k-1} + 3^k = \frac{1}{2}(3^{k+1} - 1).$$

$$\text{LHS } 1 + 3 + 9 + \dots + 3^{k-1} + 3^k$$

$$= \frac{1}{2}(3^k - 1) + 3^k = 3^k \left(\frac{1}{2} + 1 \right) - \frac{1}{2}$$

$$= 3^k \times \frac{3}{2} - \frac{1}{2} = \frac{1}{2} [3^{k+1} - 1]$$

= RHS

step 4

Since true for $n=1$, by PMI, true for all integers $n \geq 1$